YEAR TEN REVISION
GENETICS AND GENETIC ENGINEERING

1. Write the word that best describes the statement on the left side

(a) Reproduction requiring only one parent _____
(b) Reproduction requiring two parents _____
(c) Short length of chromosome (DNA) which determines a particular characteristic _____
(d) Number of chromosomes in a normal human cell _____
(e) Number of chromosomes in a human sperm or egg cell _____
(f) When a gene is “masked” or “hidden” _____
(g) Reproduction which produces offspring which are not identical to the parent(s) _____
(h) When a gene “masks” or “hides” an alternative gene _____
(i) Different genes for the same characteristic _____
(j) Identical genes for the same characteristic _____

(5 marks)

2. In Foxes, the gene for red coat colour **(R)** is dominant to the gene for silver-black colour (**r**). Determine the genotype and phenotype percentages/ratios expected for the following crosses:



(a) silver-black coat male with a heterozygous red female

(3 marks)

3. In human beings, the gene for brown coloured eyes is dominant to the gene for blue coloured eyes. A blue–eyed man marries a brown-eyed female and they have three children. Two of the children have brown eyes and one has blue eyes.



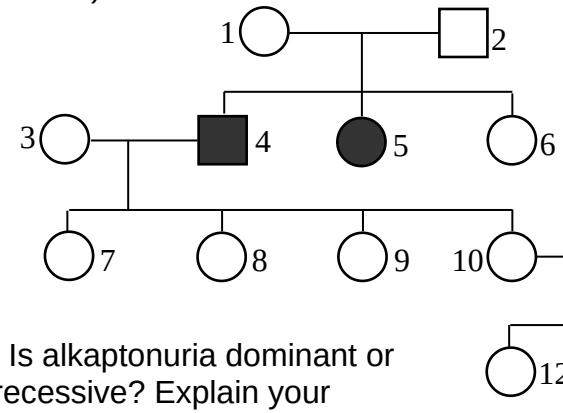
(a) What are the genotypes of the mother and the father?
(b) What is the chance that their next child will have blue eyes?

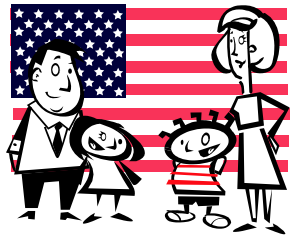
(3 marks)

4. A boy wanted to work out what his blood type would be. He knew his father had blood type **AB** and his mother had **O**. What blood types could he possibly be? Explain your answer.

(3 marks)

5. Alkaptonuria or Maple Syrup disease is a rare genetically inherited disease found in humans. In this disease the urine contains a substance which gives it a dark brown colour which turns black on exposure to air. The family tree below shows a family containing some individuals who have Alkaptonuria (shaded).





(a) Is alkaptonuria dominant or recessive? Explain your reasoning.
(b) Determine the genotype of individuals 1, 4, 10, and 11. Explain your reasoning for each answer.

(5 marks)

6. Colour blindness in humans is a **sex-linked** characteristic. A dad who was colour-blind and a mum who was a carrier for colour-blindness had 2 children, one boy and one girl.

(a) What is the chance the boy will be colour-blind?
(b) What is the chance the girl will be colour-blind?

(4 marks)

7. Suppose a white, straight haired guinea pig mates with a brown, curly haired animal. All five babies in their first litter have brown fur, but three are curly and two have straight hair. The second litter consists of six more offspring, all with brown fur – but two are curly and four are straight haired.
Assuming curly is dominant to straight, what are the genotypes of the parents and the offspring?

(5 marks)